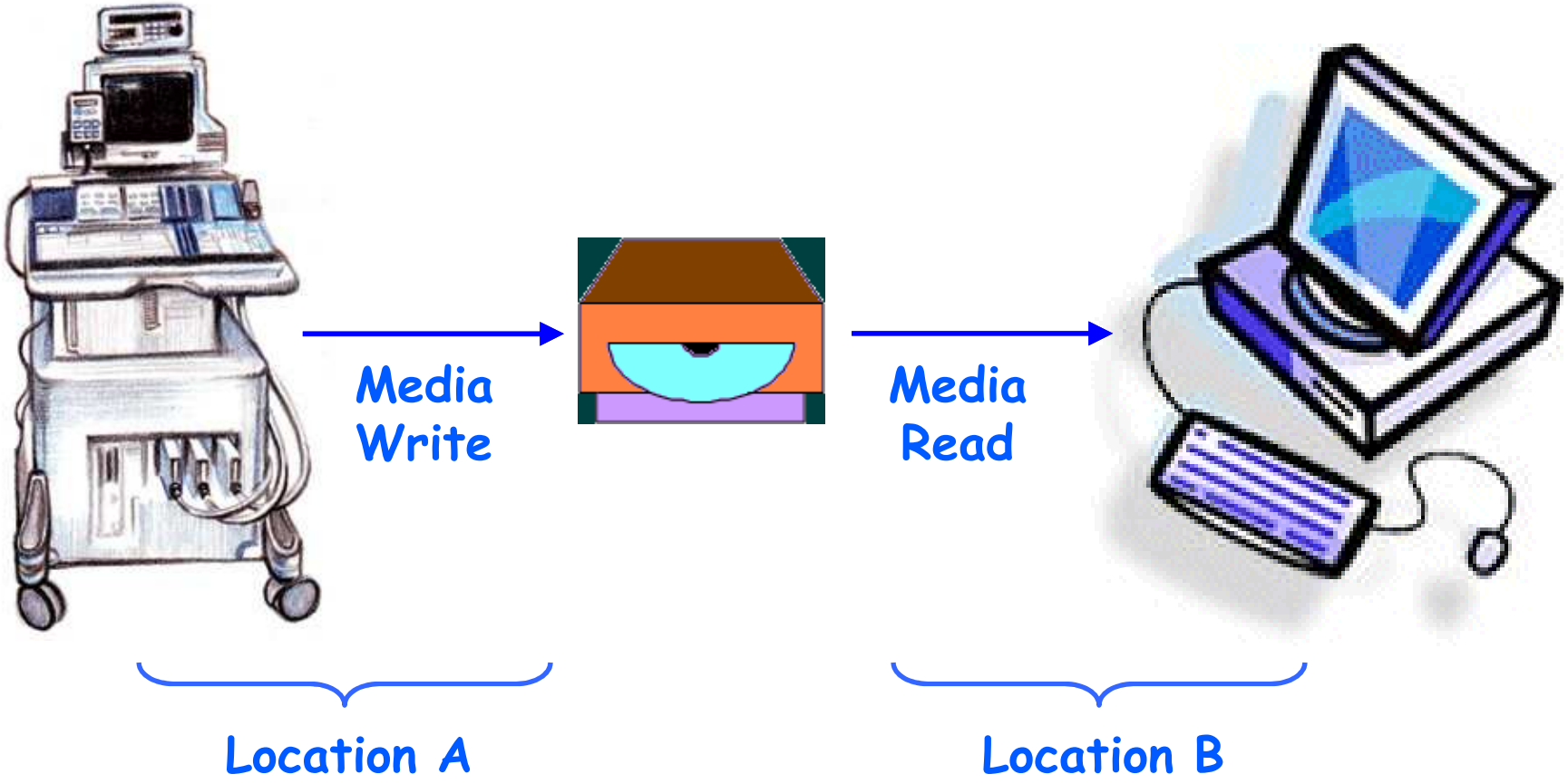
PHILIPS

DICOM Media Service

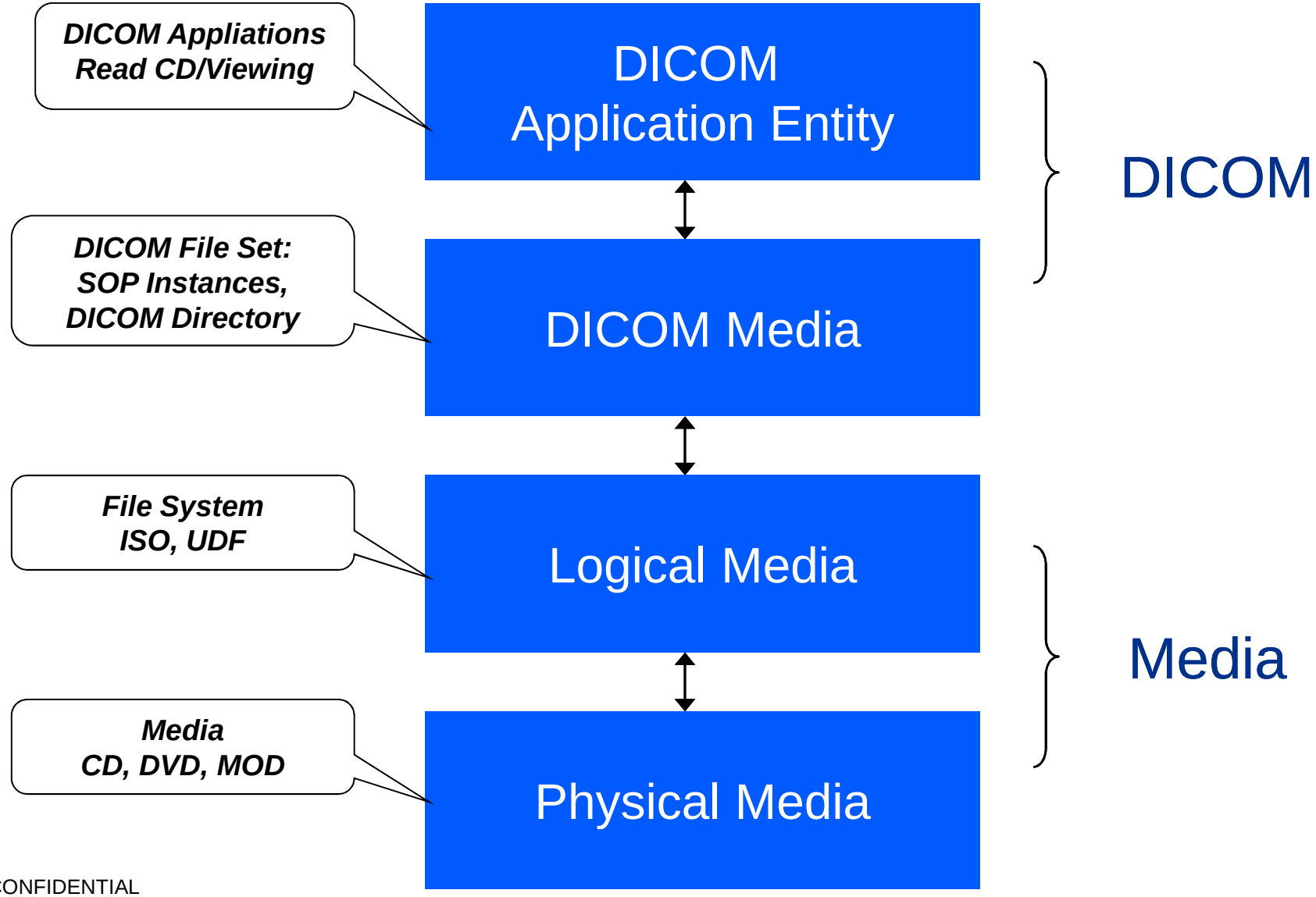
Agenda

- Need for Media Exchange
- Media Storage Model
- Media Service
- Application Profiles

Need for Media Exchange

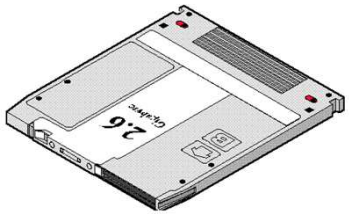


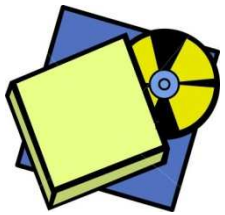
Media Storage Model



Physical & Logical Media Supported

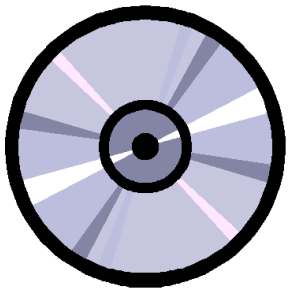
Medium Types	File System	Description
1.44 MB Diskette	FAT (12 / 16 bit) 8.3 format	Retired
130 mm 2.3 GB MOD	FAT (12 / 16 bit) 8.3 format	Retired
130 mm 4.7 GB MOD	FAT (I12 /16 bit) 8.3 format	One File set each side
120 mm CD-R	ISO 9660 level – 1 file system, 8.3 format	Only one file set allowed. Implementer may support multi-session.
120 mm DVD-RAM	UDF 1.5	One file set each side





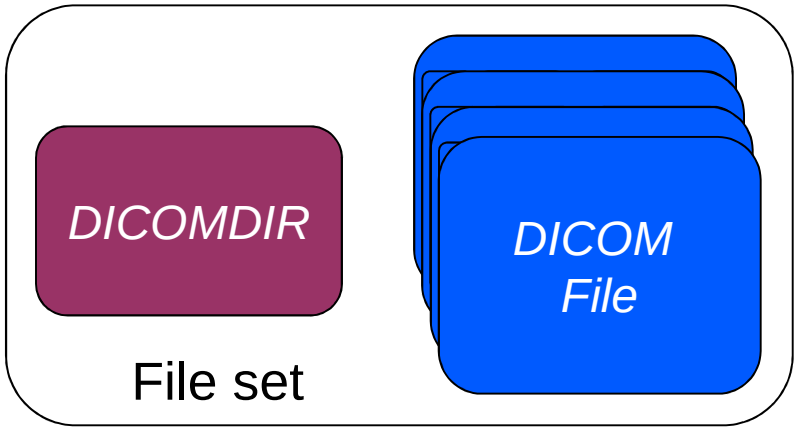
Physical & Logical Media Supported

Medium Types	File System	Description
120 mm DVD-ROM, DVD-RW, DVD+RW, DVD-R & DVD+R	UDF 1.5 ISO 9660	One file set each side
USB connected devices	FAT 16	
Through E-mail	MIME	



DICOM Media

- DICOM File
  - One DICOM SOP Instance encoded as a File
    - Known as DICOM File / DICOM Part 10 File
  - One Media Storage Directory(one or more DICOM Files)
    - Known as DICOMDIR
- Also known as DICOM File Set.



DICOM File/DICOMDIR

- First 128 byte File Preamble
- 4 byte DICOM prefix-DICM
- Meta Header (encoded in Explicit VR Little Endian(1.2.840.10008.1.2.1))
- DICOM Data

DICOM File

DICOM Meta Header

Patient
- Patient ID
- Patient Name
- Patient Sex

Study
- Study ID
- Study Date
- Study Time
- Study UID

Series
- Modality
- Series Date
- Body Part
- Series UID

Image
- SOP ClassUID
- SOPInstance UID
- Rows
- Columns

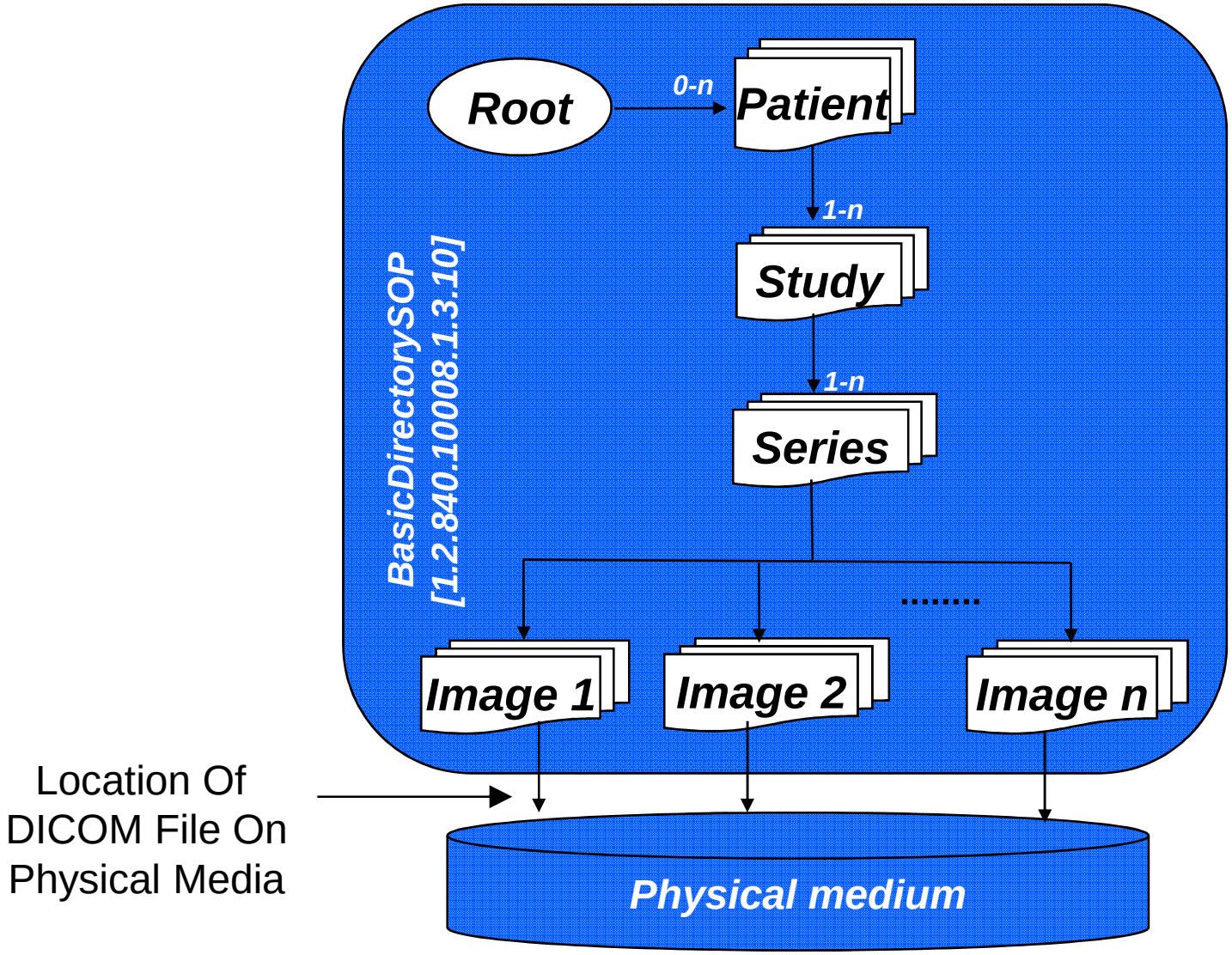
DICOM Meta Header

- Abstract content of the DICOM File
- Contains
  - Identification for DICOM
  - Transfer Syntax Encoded
    - Default is Implicit VR Little Endian
  - SOP Class Stored in the File
  - File Set ID to which the File Belongs to

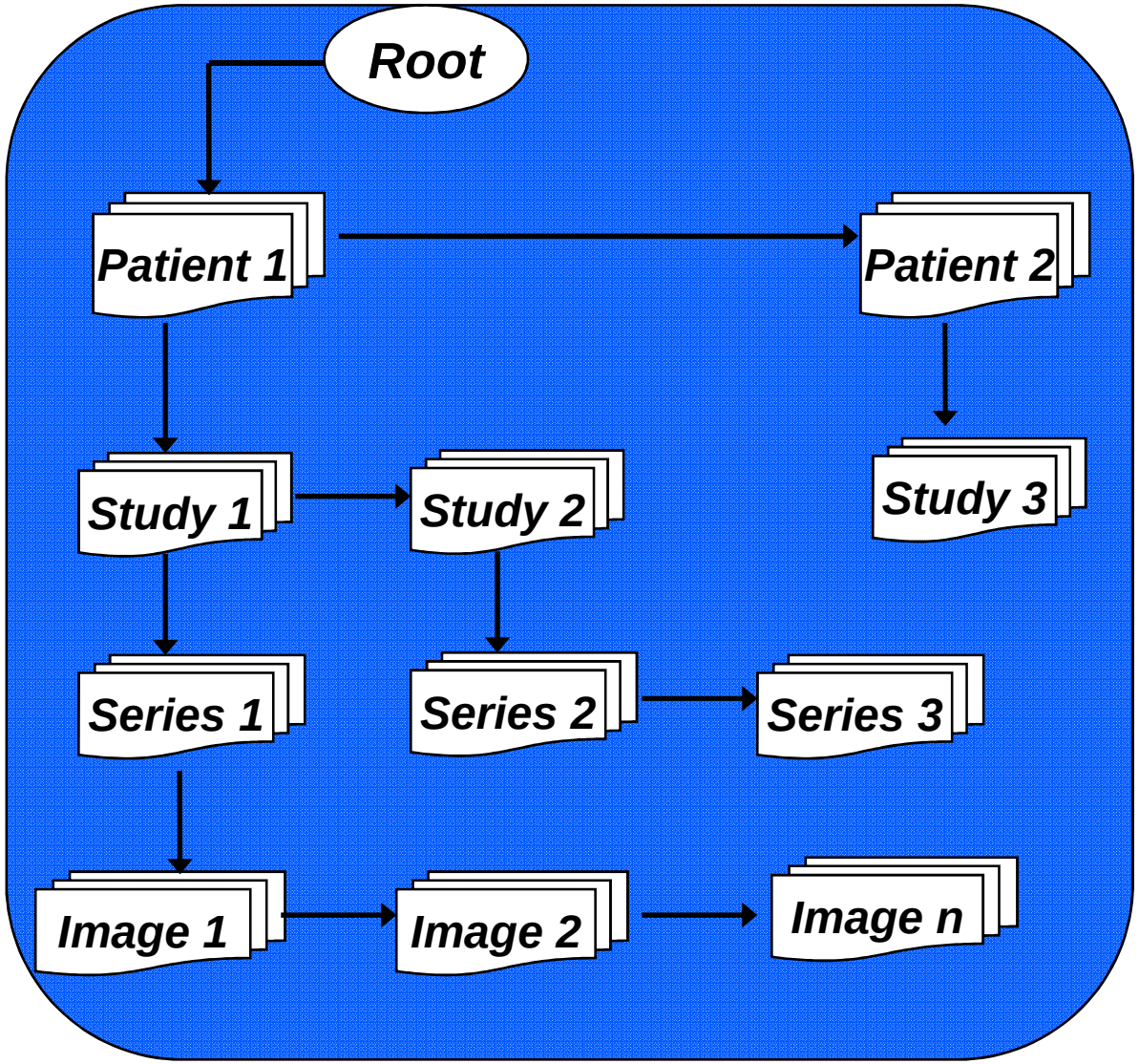
DICOM DIR

- Directory Index for all the DICOM Files
- Contains
  - Snapshot of the Information Represented by the DICOM Files
  - Information stored in Hierarchical representation
  - The relative location of all DICOM Files
- Used for Querying the Contents on the Media

DICOM DIR


Root
0-n
Patient
1-n
Study
1-n
Series
BasicDirectorySOP
[1.2.840.10008.1.3.10]
.....
Image 1
Image 2
Image n
Location Of
DICOM File On
Physical Media
Physical medium

DICOM DIR


Root
Patient 1
Patient 2
Study 1
Study 2
Study 3
Series 1
Series 2
Series 3
Image 1
Image 2
Image n

DICOM File Service

- Three Roles for DICOM File Service
  - File Set Creator (FSC)
    - Can Create DICOMDIR and DICOM Files
  - File Set Reader (FSR)
    - Can access DICOMDIR and DICOM Files
  - File Set Updater (FSU)
    - Can Read DICOM Files and can Read and Update DICOMDIR

DICOM Application Profile

- Also referred to as the Media Application Profile
  - Description of the Application
  - Media Format and Physical Medium
  - File Service Supported (FSC/FSR/FSU)

DICOM Media Service

- Demo
  - DICOM File
  - DICOMDIR

Exercise

- Draw the Tree Representation of DICOM DIR from the DICOM Dump provided

PHILIPS

sense and simplicity